

The Red/Black Game III:

Belief Carrying, Safe Re-Solving, and the Certificate Problem

(A Program With Measured Intermediate Results)

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Abstract

Paper II left heads-up Red/Black in a precisely known, uncomfortable place: the belief-reset abstraction G' is solved exactly, and the solved profile's real-game exploitability is certified to be *at least* 0.598 per game — with no upper bound of any kind. This paper is about the upper bound. We first show that carrying beliefs closes the measured gap: re-solving each round boundary under the exact public posterior (tracked by a filter over Paper I's rank-one belief family, validated to 10^{-12}) produces a complete dose-response, $\varepsilon_G \geq 0.598$ with no carrying falling to ≥ 0.013 $[-0.10, +0.13]$ with full carrying — as far as our strongest probe can see, the belief reset was the *entire* measured exploitability. But a probe is a lower bound, and naive re-solving famously admits adversarial counterexamples, so the second half builds the certification machinery: a safe-resolving gadget whose per-resolve *margins* are measured live by exact best-response sweeps, and a composition theorem — every constant re-derived by script — bounding the agent's exploitability by the root residual plus the worst reachable chain of margins, with no opponent model anywhere. We report the program's current, audited status with the candor it requires: the theorem is sound; the filter, gadget, frontier libraries, and ledger all exist and are gated; and the certified upper bound today is the trivial one, because fixed promise surfaces are *certifiably* dishonorable (their family-supremum dishonor is ≈ 1.0 at every measured layer even where on-path margins measure 3×10^{-4}). The sandwich stands at $[0.013, 2]$. We locate the entire remaining distance in one quantity — honorability of the price surface at layers $T' \geq 4$ — and lay out the four-item program that targets it, each item with the measurement that motivates it. Every number in this paper is produced by scripts in the accompanying repository.

1 The problem: an upper bound on real-game exploitability

Papers I and II established, in order: the real game G cannot be tabulated (2.4×10^{33} infosets); its reachable beliefs live in the rank-one family $\lambda_\kappa \cdot g_0 \cdot g_1$ (Paper I, Thm. 11.3); the abstraction G' that discards the tilts is exactly solvable; and discarding the tilts costs at least 0.598 per game of real exploitability, certified by a probe that wins actual games (Paper II, Sec. 9.8). The project's remaining question is the one no lower bound can answer:

Exhibit an agent for heads-up Red/Black together with a proof that no opponent strategy beats it by more than ε_G , for a small, computed ε_G .

Three disciplines frame everything below.

The sandwich. Every claim lives between two jaws: probe lower bounds (Paper II’s LBR family, extended here with resolver-aware variants whose opponent model is byte-exact) and the certificate upper bound. The jaws must never cross; a certificate below a valid probe reading refutes the certificate, not the probe. We report both jaws in every results table, and the honest summary of this paper is that the lower jaw sits at 0.013 and the upper jaw at the trivial cap.

Certification without an opponent model. A bound that assumes the opponent plays any particular way — including “the way our stored profile plays” — is not a bound. Every certified quantity below is either computed offline over all opponent types, or measured live by an *exact best-response sweep* that includes the opponent’s certified alternative.

Ship gate. $\varepsilon_G \leq 0.10$ per game (5% of the value range), stretch 0.05; secondary gates: sandwich consistency, defense non-regression against the unsafe prototype, head-to-head and bot-battery parity per Paper II’s standing battery.

2 The belief state and the filter

The agent’s belief plumbing is a direct application of Paper I’s structure theorems, and its correctness burden is discharged once, here, for everything downstream.

The filter. At public history h ending at a boundary of class κ , the exact joint posterior over type pairs is $\mu_h \propto \lambda_\kappa \cdot g_0 \cdot g_1$, and the filter tracks the two tilt vectors by per-round transfer sums: each seat’s factor is updated by the likelihood of the public record under that seat’s strategy, with the deal coupling riding along unchanged (Paper I, Lemma 11.2: $a = r + x$ is invariant). Validation: against an independent brute-force replay of all consistent histories, the filter reproduces the exact posterior to 10^{-12} ; the 2,000-game TV measurement of Paper II cost 64 seconds *including* play, so exact belief tracking is computationally free at this game’s scale.

Deviation conditioning. A re-solving agent deviates from the stored profile, so its *own* transfer must use its actually played strategies (the filter accepts a per-seat solution override); the opponent’s transfer uses the stored profile as a model. The byte-exact shadow gate (Section 4) pins this: replaying a full game’s public record through the filter reproduces the resolver’s tables bit-for-bit.

The conditionals that cannot be wrong. The opponent-side transfer is a *model*; a guarantee cannot lean on it. It does not have to:

Lemma 2.1 (Model-free conditionals). *For any opponent strategy τ and public history h , the conditional distribution of the agent’s type given the opponent’s type is $\Pr(s \mid t, h) \propto K^{\text{deal}}(a(s), a(t)) L_A(s)$, where L_A is the agent’s own likelihood sum. It does not depend on τ , and the filter update preserves it under any strictly positive opponent weighting, because a -invariance makes the deal kernel constant along both seats’ transitions.*

Proof. The deal couples the seats only through (a_0, a_1) , deterministic invariants of the current types. Path masses factorize per seat given the coupling, and the opponent’s factor multiplies whole columns, cancelling from every conditional. Every own transition preserves $a(s)$ and

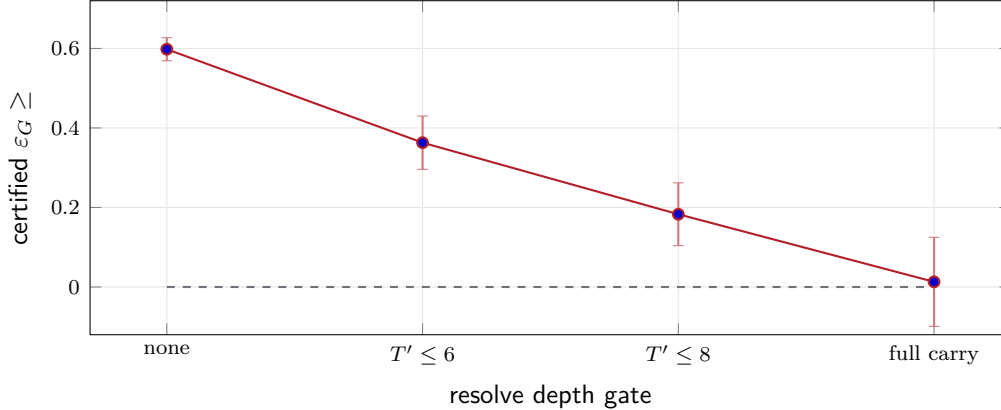


Figure 1: The dose-response of belief carrying. Each point is a certified lower bound on ε_G from the probe family (mirrored seeds, Wilson 95% intervals): 0.598 with no carrying (the stored profile, Paper II), 0.363 resolving only boundaries at $T' \leq 6$ (0.97s/resolve), 0.183 at $T' \leq 8$ (3.3s/resolve), and 0.013 $[-0.10, +0.13]$ with full carrying (≈ 4.9 s/resolve, 5.8 resolves/game). As far as this instrument can see, the belief reset was the whole of the measured exploitability.

every opponent transition preserves $a(t)$, so the kernel factor pulls out of both transfer sums unchanged. Gate: support-only opponent transfers reproduce the exact filter’s conditionals to 10^{-10} on played traces (`test_support_filter_conditionals`). $\square$

Two measured warnings sharpen the lemma’s value. The tilts are *not* the seats’ standalone likelihood products — the layer prior is a different kernel at every class, and naive per-seat products miss the kernel drift by TV up to 0.129 on played traces. And a zero-likelihood opponent trace (an opponent the stored model thinks impossible) would break a naive filter; the safe agent instead swaps the opponent transfer for its structural support indicator and keeps resolving, which by Lemma 2.1 preserves every conditional the certificate consumes. Robust belief tracking against arbitrary opponents is a *construction*, not a hope.

3 Unsafe re-solving: the gap closes, and why that is not a proof

The prototype changes exactly one input of Paper II’s machinery: at a boundary with public history h , solve the entry class’s one-round subgame with entry weights μ_h instead of the class prior λ_κ , play one round from the resolved strategy, repeat. Successor leaves keep the stored values; the kernel, the class DAG, and the row structure are untouched.

Gates before numbers. A prior-weight resolve reproduces the stored solution (the change degenerates correctly); shadow replay reproduces the resolver’s tables byte-exactly from the public record; full games run with zero fallbacks. Probing a deterministic resolver requires the probe’s opponent model to *be* the resolver: the resolver-aware probe replays a deterministic shadow copy, gated to byte-exactness, so both sides of every comparison hold equally exact models.

The two findings that define the generation. First, belief carrying gains *nothing* head-to-head: the resolver beats the stored profile 50.3% $[47.2, 53.4]$ — an equilibrium of the correctly-

believed subgame does not harvest a fixed opponent; only best response does. Second, it is a large *defensive* win (Figure 1). Beliefs pay in defense, not offense. The cost anti-correlates with the benefit: the deep layers where TV is worst (0.92–0.95) are precisely the cheapest to resolve (≤ 1 s), which is what makes full carrying practical.

Why none of this certifies anything. Three obstructions, each fatal alone:

1. **Unsafe re-solving has no theorem.** Each resolve derives μ_h assuming the agent’s own past strategy; an opponent who knows this can deviate early to poison the assumption and cash in later. Known constructions in the re-solving literature make naive re-solving strictly *more* exploitable. Our probe found no such attack — which is evidence about the probe.
2. **The filter’s opponent transfer is a model** (Section 2); a guarantee must hold against opponents no filter models.
3. **The leaves are G' -flavored.** Every resolve prices its exits with stored values computed under class priors at deeper boundaries; their error *as G -values* is the dominant unknown — and Section 6 measures it as substantial (0.146 worst-case at $T'=3$).

The structural inequality $\varepsilon_G \leq \varepsilon_{\text{comp}} + 2U \cdot \mathbb{E}[\text{TV}] \cdot \text{depth}$ is vacuous at $\text{TV} = 0.842$ (right side ≈ 10 against a trivial cap of 2) and does not touch obstruction 1 at all. So the certificate requires a different agent — the one the theorem of Section 5 is about.

4 The safe-resolving gadget

Fix the agent seat p and opponent q . All values are seat-0 payoffs in $[-1, 1]$; $U = 2$ is the payoff range. A boundary b is a full public history ending at a procurement resolution; $\kappa(b)$ its class, g_b the agent’s exact own tilt there, K_b the class’s deal-coupling kernel. The agent plays the stored root solve at the game root, and at every boundary it reaches, the strategy extracted from the gadget solve below; the *leaf functional* $\hat{v}_b(b', s', t') \in [-1, 1]$ prices round-end leaves (currently the stored values; Section 6 upgrades them), and terminal leaves are exact.

Definition 4.1 (Safe-resolving gadget). At boundary b with own range g_b and any strictly positive weighting ν_b on the opponent’s types, the gadget game $\tilde{G}(b)$ is: chance draws (s, t) with weight $K_b(s, t) g_b(s) \nu_b(t)$; the opponent observes t and chooses OUT, ending the game at the *promise* $w_b(s, t)$, or IN, after which the one-round subgame of $\kappa(b)$ is played with leaves priced by $\hat{v}_b$. The agent never opts out.

Define the conditional promise $\hat{w}_b(t) := \sum_s K_b(s, t) g_b(s) w_b(s, t) / M_b(t)$ with $M_b(t) := \sum_s K_b(s, t) g_b(s)$, and for any agent round strategy σ and opponent behavior τ the opponent’s conditional counterfactual value $\tilde{V}_q^\tau(b, t; \sigma)$ in the gadget. The weighting ν_b cancels from every conditional quantity; it tunes convergence only.

Lemma 4.2 (Margin identity). *Let σ_b be the agent’s average strategy from the gadget solve and $m_b(t) := \max_\tau \tilde{V}_q^\tau(b, t; \sigma_b) - \hat{w}_b(t)$ its measured per-type margin (one exact best-response sweep, recorded live). Then $m_b(t) \geq 0$, and every opponent behavior in the gadget satisfies $\tilde{V}_q^\tau(b, t; \sigma_b) \leq \hat{w}_b(t) + m_b(t)$.*

Proof. The best response includes OUT, whose conditional value is exactly $\hat{w}_b(t)$, so the maximum is at least $\hat{w}_b(t)$; the inequality is the definition of the maximum. The sweep is exact: the opponent’s in-round information sets are (public node, private state) pairs, swept by per-state argmax against path-summed agent reach, then a per-type maximum over pre-stage choices including OUT. $\square$

Lemma 4.3 (Carry-down consistency). *Construct promises and prices from one functional $\hat{v}$: at every resolve, $w_b := \hat{v}$ on the entry grid of $\kappa(b)$, and leaves priced by $\hat{v}_b$. If b' is a successor boundary reached from the resolve at b , with $g_{b'}$ the played-strategy transfer of g_b , then the conditional promise of the next resolve equals the conditional leaf price the b -resolve charged: $\hat{w}_{b'}(t') = \sum_{s'} K_{b'} g_{b'} \hat{v}_b(b', s', t') / M_{b'}(t')$.*

Proof. Both sides are the same $g_{b'}$ -weighted average of the same matrix: the filter propagates its own factor under exactly the played strategy (the byte-exact shadow gate), and the opponent factor cancels from the conditional by Lemma 2.1. Verified as a float-precision identity on played traces. $\square$

Remark 4.4 (One-sidedness, and why it matters). Lemma 4.3 is an equality for the one-functional construction, but the telescoping argument below survives promises *at or below* the parent’s leaf conditionals — a promise may be tightened downward without breaking soundness. This freedom is what the “hybrid” promise experiments of Section 7 price, and its measured failure mode (tightening without enforcement is harmful) is one of the program’s sharpest lessons.

Implementation status, all gates green: prior-weight gadget solve reproduces the stored value (margins under $20\times$ solver tolerance); the compiled kernel and the reference implementation agree on the gadget game to 10^{-9} at identical iteration counts; safe shadow replay is byte-exact (10^{-12}); full games run with zero fallbacks, including forced filter-break survival; and the gadget’s conservatism costs -1.4 percentage points head-to-head against the unsafe resolver (CIs overlapping) at $\varepsilon_{\max} = 8.9 \times 10^{-4}$ per resolve.

5 The composition theorem

Assumption 5.1. Full carry: every non-terminal boundary the agent reaches is resolved. A depth-gated agent adds, per unresolved round, an uncertified belief-reset term; the certificate is a full-carry object.

Assumption 5.2. No unrepaired filter break — enforced by construction in safe mode (Section 2); the repair preserves every conditional the certificate consumes and cannot itself break.

For opponent type t at reachable boundary b , let $\text{CBV}_A(b, t)$ be the supremum over opponent play below b of the opponent’s true conditional counterfactual value in G against the agent’s continuation, with the $K_b(\cdot, t) g_b$ counterfactual weighting. Reachable chains are sequences $(b_1, t_1) \rightarrow \dots \rightarrow (b_L, t_L)$ of successor boundaries and opponent types under the agent’s public dynamics; a game has at most $L_{\max} = 8$ resolved boundaries ($T' = 9, \dots, 2$; re-derived by script).

Theorem 5.3 (Composition, measured form). *Under Assumptions 5.1 and 5.2, for every opponent strategy τ :*

$$u_q(A, \tau) \leq B_q^{\text{root}} + \max_{(b_1, t_1) \rightarrow \dots \rightarrow (b_L, t_L)} \sum_{i=1}^L m_{b_i}(t_i),$$

where B_q^{root} is the opponent's exact best-response value against the stored root tables in the $\hat{v}$ -priced root round, and the maximum ranges over reachable chains. For the mirrored agent pair, $\varepsilon_{\text{pair}} \leq \varepsilon_{\text{round}}(\text{root}) + \sum_{\text{seats}} \max_{\text{chains}} \sum_i m_{b_i}(t_i)$.

Proof. Bottom-up induction on boundaries with $D(b, t) := m_b(t) + \max_{(b', t') \text{ reachable}} D(b', t')$ (empty max = 0); the claim is $\text{CBV}_A(b, t) \leq \hat{w}_b(t) + D(b, t)$.

Base: a resolve whose leaves are all terminal has $\hat{v}$ exact, so the gadget is the truth below b restricted to IN, and Lemma 4.2 closes it.

Step: fix τ below b . The true conditional value decomposes over the round's leaf aggregates: writing ρ_τ for opponent-and-chance reach into the aggregate landing on (b', t') ,

$$V_q^\tau(b, t) = \tilde{V}_q^\tau(b, t; \sigma_b) + \sum_{(b', t')} \rho_\tau \left[V_q^\tau(b', t') - \frac{\sum_{s'} K_{b'} g_{b'} \hat{v}_b(b', s', t')}{M_{b'}(t')} \right],$$

because gadget and truth differ only in leaf pricing, and the agent's reach into each leaf under σ_b is exactly what defines $g_{b'}$. By Lemma 4.3 the subtracted term is $\hat{w}_{b'}(t')$; by induction the bracket is at most $D(b', t')$; by Lemma 4.2 the first term is at most $\hat{w}_b(t) + m_b(t)$. Take suprema.

Root: the same decomposition applied once to the stored root round, prior-weighted over root types, plus the zero-sum meter identity $B_0^{\text{root}} + B_1^{\text{root}} = \varepsilon_{\text{round}}(\text{root})$. $\square$

Remark 5.4. The bound is opponent-model-free: the weighting ν_b and the filtered opponent factor influence which equilibrium the resolve finds — hence how small the margins come out — never the validity of the theorem. This is exactly the property unsafe re-solving lacks.

End-to-end verification: the composed two-level agent (gadget resolve at a $T'=3$ boundary plus per-history gadget resolves at its $T'=2$ successors) is transcribed into the flattened perfect-recall oracle, the opponent is exact-best-responded per type, and the result is checked against $\hat{w}$ plus the chain-sum of recorded margins — 5/5 configurations pass (both seats, prior and tilted entries, including the worst promise-dishonor class); carry-down consistency holds to 3×10^{-16} .

6 The exact frontier

The theorem charges margins; margins are small only where promises are *honorable* — backed by something true in G . The stored values are not: at $T'=3$, 37 of 66 class values sit outside their exact-in- G brackets, worst case 0.146 (stored -0.657 against exact -0.510 $[-0.5105, -0.5100]$), reach-weighted 1.36×10^{-2} — the within-subgame belief-reset composition error, in agreement with the full-carry probe residual (0.013), a satisfying cross-instrument consistency. So the certificate needs a new object: exact values of class subgames *as functions of entry belief*.

The domain correction. $V(\kappa, \mu)$ is not a function on the full belief simplex and is not piecewise-linear there. By Paper I's factorization the reachable domain is the rank-one family, and restricted to it V is piecewise-*bilinear* in the tilt pair (g_0, g_1) : concave piecewise-linear in each tilt with the other fixed. A facet census confirmed the model: across sampled seat segments the worst midpoint-above-chord excess is negative (every midpoint at least 1.4×10^{-3} below chord plus certified slack).

The facet census. The operational complexity measure is the size of a greedy *strategy-library cover*: solved round strategies such that exact best-response sandwiches (lower and upper envelopes over library entries) close to 10^{-3} over a set of candidate beliefs. Full population at $T' \leq 3$:

	$T' = 2$	$T' = 3$
classes (canonical, pending-discard)	18	66
library size min / median / max	1 / 29 / 40	1 / 28.5 / 47
residual cover gap (max)	8.1×10^{-4}	9.9×10^{-4}
holdout gap median / max	2.3×10^{-3} / 0.196	4.5×10^{-3} / 0.657
stored value outside exact bracket	0/18	37/66
stored-minus-exact delta, median / max	1.2×10^{-4} / 2.0×10^{-4}	4.3×10^{-4} / 0.146

Facet counts are tens per class against a kill criterion of 10^5 : the exact frontier is cheap where it was feared expensive. $T'=2$ stored values are exact in G (18/18 inside brackets) — the layer where $G' = G$, confirmed. Covers are dense where mass goes and weak at belief-space vertices (holdout max 0.657), which returns with force below. At $T'=4$, buildability and reach *anti-correlate*: the most-visited classes exceed flat-tree build budgets, so deep-layer certification leans on functional pricing at $T'=5$ boundaries rather than exact $T'=4$ leaves for the heavy classes. Frontier validation is oracle-, mirror- and $T'=2$ -containment-based; stored-value equality is *not* a gate above $T'=2$, by the measurement above.

What the frontier owes the certificate. The theorem quantifies over all opponents, including ones steering to boundaries no probe visits, so the certificate needs an a-priori bound on $m_b(t)$ valid at every reachable (b, g_b) . Per gated class κ and every member g of its factored entry family, items in force: (1) an *honorable promise vector* $w_\kappa(g, \cdot)$ — a certified guarantee that some playable library strategy caps the opponent’s true conditional CFV below the boundary at $w_\kappa(g, t) + r_\kappa(g, t)$ for all t simultaneously, with certified residual r ; (2) the *functional sandwich gap* $\Delta_\kappa(g, t)$ between the library’s envelopes evaluated at the resolve’s own entry ranges (two exact-BR sweeps per entry, ~ 5 ms) — the leaf-error norm of record; (3) an *off-path enforcement rule*: the resolver compares its solve’s measured margins against the capping entry’s and plays the better, converting measured margins into the a-priori form $m_b(t) \leq \varepsilon_b + r_{\kappa(b)}(g_b, t)$.

Corollary 6.1 (Certified form). *With (1)–(3) in place, $\varepsilon_{\text{pair}} \leq \varepsilon_{\text{round}}(\text{root}) + \sum_{\text{seats}} \max_{\text{chains}} \sum_{i=1}^{L \leq L_{\text{max}}} [\varepsilon_{b_i} + r_{\kappa(b_i)}(g_{b_i}, t_i)]$, every term computed offline or recorded live.*

7 The ledger, and the audited verdict

Everything above exists in code with green gates. This section reports what happens when it is all assembled — kept deliberately close to the project’s internal ledger, because the assembly is where an earlier draft of this program fooled itself, and the audit that caught it is part of the result.

One functional, measured margins. With the one-functional carry-down (a single price surface: frontier prior-entry values where covered, stored values elsewhere — used as both leaf pricing and promises), Lemma 4.3 holds by construction at every layer and the meter isolates exactly one quantity: honorability of the price surface.

Table 1: Per-layer certificate terms under the one-functional carry-down (171 resolves over 32 full-carry games): mean-of-max per-type margin plus mean resolve residual ε_b . The mechanism proves itself at $T'=2$, where promises are exact in G and the meter vanishes; the distance is the stored surface at $T' \geq 4$.

Layer	margin (mean-of-max / max)	ε_b (mean)	price surface
$T'=9$	0.134 / 0.363	9.3×10^{-4}	stored
$T'=8$	0.133 / 0.734	9.1×10^{-4}	stored
$T'=7$	0.111 / 0.508	8.7×10^{-4}	stored
$T'=6$	0.106 / 0.310	2.1×10^{-4}	stored
$T'=5$	0.148 / 0.900	2.4×10^{-4}	stored
$T'=4$	0.258 / 0.790	9.1×10^{-5}	stored (libraries landing)
$T'=3$	0.085 / 0.499	9.6×10^{-5}	frontier (prior point)
$T'=2$	0.0003 / 0.0006	1.7×10^{-4}	frontier — <i>honorable</i>

Table 1 carries the whole story. Where the surface is honorable ($T'=2$: terminal-exact frontier values) the margin term collapses to gap scale — the certificate mechanism works. Where it is the stored surface ($T' \geq 4$) each resolve charges 0.11–0.26, and layers 4–9 carry 91% of the per-seat chain. Resolve residuals (10^{-4} scale) are irrelevant everywhere: there is nothing to buy by solving harder.

Incident — the 1.958 that was not a bound (2026-08-03)

Summing Table 1’s on-path terms over $L_{\max} = 8$ and both seats gives 1.958 — and an earlier draft of this ledger reported it as the certificate. An independent adversarial audit rejected it: the theorem’s chain maximum ranges over *opponent-steerable* chains, and an on-path mean substitutes a self-play average for a supremum the opponent controls. The licensed assembly from the same data — worst observed chain over visited boundaries only — is 8.22: vacuous. Reach-weighting is not a sound repair (the opponent steers the carried beliefs; self-play samples a measure-zero slice of the chain space). The audit’s verdict, adopted in full: **certified ε_G today is the trivial cap**. The sound averaging opportunities that survive — root-prior weighting over the opponent’s root type, and a chance-expectation reproof of the successor step (the verification’s 0.4–2.0 slack locates the loose inequality) — are theorem work, listed in the program. Lesson: in certification, the assembly is as much a proof obligation as the per-piece inequalities, and an average is never a supremum.

Fixed promise surfaces are certifiably dead. The audit also produced the program’s most decisive positive tool. Per-type dishonor of a *fixed* promise surface is linear-fractional in the own tilt, so its supremum over the rank-one family is attained at own-type vertices and can be certified exactly (a clairvoyant best response per vertex). Result: the func surface’s certified sup dishonor is median ≈ 1.0 , max 2.0, at *both* $T'=2$ and $T'=3$ — even where on-path margins measure 3×10^{-4} . An earlier holdout measurement suggesting fixed refined surfaces might suffice (max 0.063) was a sampling artifact of insufficiently extreme tilts. Any certificate path through a fixed promise matrix is closed, permanently, at every layer; what remains open is *range-dependent* promises (library caps evaluated at the realized entry range) with per-type enforcement, whose residual is the library sandwich-gap supremum — itself lower-bounded by vertex probes at 0.14–0.43 median today, hence the vertex-completion item in the program.

Enforcement is mandatory, not optional. Honorable (tighter) promises push opponent opt-out mass up, and CFR’s weighted objective then simply ignores opted-out types — their margins are only weakly controlled by the solve. Measured: hybrid promises (tightened by Remark 4.4) *worsen* the $T'=3$ margin term from 0.085 to 0.634 as built, because the meter measures against the tighter bar while the solve stops defending it. The audit’s a-priori judgment (“sound but worthless without enforcement”) was generous; the measurement says actively harmful. Post-solve per-type minimization with the capping entry — item (3) above — is load-bearing machinery.

Table 2: The certificate scoreboard. Lower bounds are probe results (mirrored seeds, $n = 300$ games, Wilson 95%); the upper bound is the audited assembly status. The ship gate is 0.10.

Instrument	value
probe LB, unsafe full carry	$\varepsilon_G \geq 0.013$ $[-0.10, +0.13]$
probe LB, safe + stored promises	$\varepsilon_G \geq 0.160$ $[0.047, 0.269]$
probe LB, safe + certified $T' \leq 3$ promises	$\varepsilon_G \geq 0.113$ $[0.000, 0.224]$
certificate UB (audited)	trivial (2.0)

The sandwich at window close. Two readings of Table 2 matter. The safe agent with dishonorable stored promises is *measurably more exploitable* (0.160) than the unsafe agent (0.013): safety machinery with bad promises is worse than no machinery, because the opponent can farm the dishonor — the probe converts exactly the margins of Table 1 into realized wins, which confirms the dishonor is substance, not accounting. And certified promises at $T' \leq 3$ move the point estimate in the right direction ($0.160 \rightarrow 0.113$) but $n = 300$ cannot separate them; the certificate’s upper bound, not more probe games, is the instrument of record — a deliberate allocation choice, since separating the variants at 95% costs 10–20k games of compute that certifies nothing.

8 The program

Four items, each with the measurement that motivates it, in dependency order. The generation’s risk is concentrated in item 2.

1. **Vertex-complete frontier libraries at $T' \leq 4$.** One exact solve per feasible own-type vertex (10–144 per class, sub-second each) collapses the library gap at vertices to solve tolerance; the interior sup then follows from the piecewise-bilinear structure by small bilinear programs. Target: certified per-class residual r_κ at $T' \leq 4$ of order 5×10^{-3} . Engineering plus modest theory; the vertex probes above say the gap is real and say exactly where.
2. **Certified cap propagation upward.** A $T' \geq 5$ cap against a strategy is one kernel exact-BR round sweep with capped leaves — sound by Theorem 5.3’s machinery. The open question is whether six layers of propagation stay tight; the measured evidence for hope is that interval propagation *contracts* per layer (0.66 – $1.03\times$, round mixing diluting leaf error) rather than fattening. This is the research risk of the generation, and it is measurement-first testable in days.

3. **Enforcement transcription.** Library entries transcribed to playable round strategies; per-type minimization at resolve time. Converts measured margins into a-priori bounds and flips the hybrid experiment from harmful to a strict win.
4. **Sound accounting.** The audit’s surviving averaging opportunities: margin de-duplication (measured -0.007), a -invariant chain assembly (per-type margin vectors are persisted for exactly this), root-prior weighting, and the chance-expectation reproof of the successor step.

Probe power is a compute line item only after items 1–3 move the certificate; the sandwich’s upper jaw is where the next result lives.

Out of scope. Serving latency (the resolve-service versus belief-bucket question), multiplayer, public-statistic enrichment (dead by Paper II’s refinement ladder), further equilibrium-set selection (real-game noise by Paper II’s measurements).

9 Verification

House rule, unchanged from Papers I–II: no number enters this paper without a script that regenerates it. `v6_verify_composition.py` re-derives $L_{\max}$, checks Lemma 4.3 as a float-precision identity on played traces, and verifies Theorem 5.3 end-to-end against the perfect-recall oracle on composed two-level agents (5/5 configurations). The filter, gadget, shadow-replay, filter-break and kernel-parity gates run in the simulator’s test suite; the facet census, fattening, margin-by-layer, vertex-sup and probe measurements each write a JSON artifact whose values appear here verbatim. The perfect-recall oracle merges procurement flip interleavings (order irrelevance, Paper I), so the verified agent variant feeds its filter order-canonicalized records; this changes nothing in the theorem, which never references flip order.

Status line, for the reader who skipped to the end: the theorem is proved, the machinery is built and gated, the lower jaw is at 0.013, the upper jaw is trivial, and the entire distance is one named quantity — honorability of the price surface at $T' \geq 4$ — with a four-item program aimed at it.